INCH-POUND

MIL-DTL-32050(OS)
25 April 2000 .
SUPERSEDING
SCD 839AS100, Rev. A
12 July 1978

DETAIL SPECIFICATION

CARTRIDGE, IMPULSE, P/N 839AS100

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the M666 Impulse Cartridge, P/N 839AS100, referred to in this document as cartridge.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents specified in Sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplements thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Division (Code 840M), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-DTL-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices

(Unless otherwise indicated, copies of DoD specifications are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

U.K. MINISTRY OF DEFENSE

AIR 2178	Specification for Manufacture and Quality Assurance of Cap, Percussion, QV5AF (VH 2 Composition Filling)

(Application for copies should be addressed to the Royal Ordnance, Euxton Lane, Chorley, Lancashire, England PR7 6AD.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

839AS100	Cartridge, Impulse

(Application for copies should be addressed to the Commanding Officer, Naval Air Technical Service Facility, 700 Robins Avenue, Philadelphia, PA 19111-5079.)

2.3 Non-Government publications and drawings. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents, which are DoD adopted, are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

ASTM E 1742	Standard Practice for Radiographic Inspection (DoD adopted)

(Application for copies of ASTMs should be sent to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959)

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AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ASQC Z1.4	Sampling Procedures and Tables for Inspection by Attributes (DoD adopted)

(DoD activities may obtain copies from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094. The private sector and other Government agencies may purchase copies from the American Society for Quality Control, P. O. Box 3005, 611 E Wisconsin Ave., Milwaukee, WI 53201-4606.)

MARTIN BAKER AIRCRAFT COMPANY, LTD. DRAWING (CAGE Code U1604)

TE –26	Vented Pressure Vessel

(Application for copies should be addressed to: Martin Baker Aircraft Company, Limited; Higher Denham, near Uxbridge; Middlesex, England; UB9 5AJ, United Kingdom.)

2.4 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supercedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2) a sample shall be subjected to first article inspection as specified in section 4.2 (see 6.3). A first article inspection shall be required if:

a. Two or more years has passed since P/N 839AS100 or similar items have been manufactured,
b. there has been a major turnover in personnel,
c. there have been major changes to the production facility or
d. there has been a transfer of production to a new facility or site.

3.2 Design and construction. The cartridge shall be manufactured to drawing 839AS100, and shall meet the requirements of this specification. Primers shall be made in accordance with specification AIR 2178. Primers shall be from lots manufactured within 2 years or recertified within 12 months from the assembly date of the cartridge lot.

3.3 Cartridge production. Only one lot of primers shall be used in the manufacture of a single lot of cartridges.

3.4 Leakage. All cartridges produced under contract shall produce no leaks greater than $1x10^{-5}$ cc/sec of air when leak tested in a dry gas leak detector as specified in 4.5.

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3.5 Radiographic inspection. All cartridges manufactured under contract shall be free of imperfections in assembly when examined radiographically as specified in 4.3.2.

3.6 Ballistic requirements. Cartridges from each first article sample and production lot sample shall meet the requirements listed below when subjected to the tests prescribed in 4.6.

3.6.1 Maximum pressure. The upper limit (UL) for maximum pressure is 2,800 psi and the lower limit (LL) is 2,300 psi when tested at 70°F.

3.6.2 Time to maximum pressure. The upper limit (UL) of the time interval from first indication of pressure to maximum pressure is 40 milliseconds. The lower limit (LL) for the time to maximum pressure is 25 milliseconds.

3.6.3 Actuation time. The upper limit (UL) for the time interval from sear removal to first indication of pressure is 9 milliseconds.

3.6.4 Time to ½ maximum pressure decrement. The upper limit (UL) for the interval from first indication of pressure to ½ maximum pressure decrement is 125 milliseconds. The lower limit (LL) is 95 milliseconds.

3.6.5 Misfire. There shall be no misfires.

4. VERIFICATION

4.1 Classification of inspections.

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection. Before entering into quantity production, an acceptable first article sample of 19 cartridges shall be prepared using the same procedures and methods proposed for manufacture of the production lot. An acceptable sample is one that has passed all the tests and examinations of 3.1 through 3.5.

4.2.1 Test methodology. Eighteen of the 19 cartridges shall be expended in the tests listed herein and one shall be retained for investigative purposes. First article sample environments, as defined in MIL-DTL-21625, shall consist of the conditioning of table I, below, of this specification. Cartridges shall be fired as described in 4.6 and shall meet the functional requirements of 3.6.

4.3 Conformance inspection.

4.3.1 Visual inspection. Visually examine all cartridges and reject those with inadequate or illegible markings, serial numbers, or color coding, and those with dents, deep scratches, split or cracked edges, chipped surfaces, or other defects which may prevent proper installation or function.

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Table I. First article tests.

Environment	Quantity
Temperature & humidity Cycling:	6
Salt spray:	6
Conditioning at –65 ° F:	3
Conditioning at +160 ° F:	3
Investigative sample:	1

4.3.2 Radiographic inspection. All cartridges shall be radiographed, in accordance with ASTM E 1742, to verify the presence, condition and proper installation of parts. The samples shall be arrayed on trays or boards in consecutive numerical order and each radiograph shall carry a permanent identification of the cartridges displayed thereon. The cartridges shall be positioned to give the most revealing cross-section of each unit. All units shall be identified with serial numbers prior to radiographic inspection. The radiographic identification shall include the complete lot number and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the plate. Defective cartridges shall be clearly identified on the plate and shall be removed. Radiographs or copies of the entire production lot shall accompany the ballistic sample to the activity conducting the production conformance test (see 6.2). Any observable imperfections shall be cause for rejection of the unit.

4.4 Lot Acceptance.

4.4.1 Sampling. Cartridges for ballistic test and investigative samples shall be randomly selected from each production lot in accordance with the requirements of ASQC Z1.4. The production lot size, excluding test samples, shall not exceed 500 and the sample size shall be in accordance with table II of this specification.

Table II. Sample sizes.

PRODUCTION LOT SIZE	BALLISTIC SAMPLE	INVESTIGATIVE SAMPLE	TOTAL LOT SAMPLE
0 – 180	15	3	18
181 – 300	20	3	23
301 – 500	25	3	28

4.5 Leakage. All cartridges shall be leak tested in a dry gas leak tester to determine conformance with 3.4. Units, which exhibit a leak rate in excess of 1 X 10^{-5} standard centimeters per second of air at a pressure differential of 1.0 atmospheres, shall be rejected.

4.6 Ballistic tests. Cartridges selected in accordance with table II, above, shall be conditioned to 70 ° F for a time period of not less than 4 hours. Units shall be fired, within 3 minutes of

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removal from conditioning, in the Martin Baker test fixture TE-26. If not fired within 3 minutes, reschedule the test and recondition for 2 hours minimum.

4.6.1 Acceptance criteria. The lot shall be acceptable with respect to functional testing if the cartridges meet the requirements of 3.6. Accept on no defects, reject on one or more.

4.6.2 Misfire. A cartridge that does not fire under the conditions specified in 4.6 shall be considered unacceptable unless the misfire is clearly attributable to faulty test equipment or procedure.

4.6.3 Retest. There shall be no retest.

4.6.4 Test failure. If a test failure is attributed to an assignable cause, excluding the test sample, the original test results shall be discarded and that test reconducted.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order, (see 6.2). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The cartridges covered by this specification are used in the MB 200-259 Gas Generator to provide gas pressure to activate the gas actuated pistons on the EA-6B aircraft Electronic CounterMeasures Officer (ECMO) ejection seats. The M666, Cartridge, Impulse Cartridge is designed to generate hot gases to actuate pistons to the other three seats to initiate the ejection sequence. This cartridge is military unique and there is no equivalent commercial application for this design.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).

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c. Whether first article inspection is required and, if so, specify the test activity (see 3.1 and 4.2).

d. Production lot size (see 4.4.1) and test activity (see 4.6).

e. Packaging requirements (see 5.1 and 6.5).

f. Items of data required for each production lot.

g. Safety requirements (see 6.4).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested (see 4.2). The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles.

6.3.1 Bids. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Ammunition and explosive safety.

6.4.1 Contractor's responsibility. The safety requirements of the "Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material" (DOD 4145.26M) are applicable when this specification is used for acquisition.

6.4.2 Government activity's responsibility. The safety precaution requirements of "Ammunition and Explosives Ashore", OP5 are applicable when this specification is used as part of a description of work to be accomplished by a government activity.

6.5 Contract packaging, packing and marking. The following requirements should be specified in the contract.

6.5.1 Packaging and packing. In addition to the following, Level A and Level C packaging and packing should be in accordance with 49 CFR 100-199. For explanation of terms , see 6.7.

6.5.1.1 Level A. Level A packaging and packing should be used for packaging and packing of all production lots for service use.

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6.5.1.1.1 Inner container. A maximum of one cartridge, with one O-ring, should be packaged in a sealed metal container, which conforms to the requirements of MIL-P-116, Method IA-5, or a sealed metal container, which will provide equal protection.

6.5.1.1.2 Outer container. Fifteen inner containers should be packed in the outer container conforming to PPP-B-621, Class 2, Style 4 or 4½ or a box conforming to PPP-B-601, Overseas Type, Style A or B or equal.

6.6 Marking.

6.6.1 Special marking. Marking of outer containers should be in accordance with CFR 49 171-199.

6.6.2 Normal marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages and shipping containers should be marked in accordance with MIL-STD-129.

Inner container marking :

NSN: 1377-00-139-5933 DODIC: M666

Nomenclature: Cartridge, Impulse

Part Number: 30003-839AS100

Lot Number: (In accordance with MIL-STD-1168)

ICC Explosive Class: C

Outer container marking :

NSN: 1377-00-139-5933 DODIC: M666

Nomenclature: Cartridge, Impulse

Part Number: 30003-839AS100

Quantity: 15

Gross weight and cube:

Lot number: (In accordance with MIL-STD-1168)

Contract or purchase order number:

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"WARNING: EXPLOSIVE POWER DEVICE, CLASS C EXPLOSIVE, HANDLE WITH CARE, KEEP FIRE AWAY"

The above information should be applied in accordance with the provisions of MIL-STD-129.

6.7 Definitions.

6.7.1 Level A packaging. The degree of preservation and packaging that will afford adequate protection against corrosion, deterioration and physical damage during shipment, handling, indeterminate storage and world wide redistribution.

6.7.2 Level A packing. The degree of packing that will afford adequate protection during shipment, handling, indeterminate storage and worldwide distribution.

6.8 Explanation of terms. For the purposes of this specification, NAVSUPINST 4030.28A should be used to clarify terms.

6.9 Subject term (key word) listing.

Aircraft Ejection seat
Emergency egress
Cartridge
Gas Generator

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Custodian:
Navy – OS

Preparing activity:
Navy – OS
(Project No. 1377-0117)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

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I RECOMMEND A CHANGE:

1. DOCUMENT NUMBER
MIL-DTL-32050(OS)

2. DOCUMENT DATE (YYYYMMDD)
20000425

3. DOCUMENT TITLE
CARTRIDGE, IMPULSE, P/N 839AS100

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*

b. ORGANIZATION

c. ADDRESS *(Include Zip Code)*

d. TELEPHONE (Include Area Code)
(1) Commercial
(2) AUTOVON *(if applicable)*

7.DATE SUBMITTED (YYYYMMDD)

8. PREPARING ACTIVITY

a. NAME
Commander, Indian Head Division, Naval Surface Warfare Center

b. TELEPHONE *Include Area Code)*
(1) Commercial 301-744-4700
(2) AUTOVON 354-4700

c. ADDRESS *(Include Zip Code)*
Standardization Team (Code 840M), 101 Strauss Avenue, Indian Head, MD 20640-5035

IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:
Defense Standardization Program Office (DLSC-LM)
8725 John J. Kingman road, Suite 2533 Ft. Belvoir, VA 22060-2533
Telephone (703) 767-6888 AUTOVON 427-6888

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